

Bhargav Hegde

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Graduate AI Engineer with 3+ years of software engineering experience, specializing in Deep Learning, Computer Vision, and Autonomous Vehicle Perception Systems

Education

State University of New York at Buffalo, Buffalo, NY Aug. 2024 – Present
Master of Science in Computer Science and Engineering
Research Associate in CAVAS Lab (Connected and Autonomous Vehicle Applications and Systems) - conducting research on real-time 3D object detection, LiDAR-based perception, and V2X cooperative perception for autonomous vehicles
JSS Academy of Technical Education, Bangalore, India Aug. 2016 – Aug. 2020
Bachelor of Engineering in Computer Science and Engineering
Initiated and **secured funding** for an IoT laboratory; conducted instructional sessions for junior students

Professional Experience

Tech Mahindra, Software Engineer Nov. 2021 – Jul. 2024
Developed automation frameworks using Robot Framework, applying machine learning-based pattern recognition for Wi-Fi RTT locationing systems (IEEE 802.11mc), achieving sub-meter accuracy through advanced packet analysis with Wireshark. Led R&D initiatives in DHCPv6 protocol testing, designed comprehensive test cases addressing stateful vs. stateless configurations, and delivered training to cross-functional teams globally on Wi-Fi RTT implementation.
Tata Consultancy Services, System Associate Engineer Apr. 2021 – Oct. 2021
Developed Python-based RESTful APIs to automate business processes, streamlining data fetching, processing, and storage for efficient client reporting, accelerating deployment time by 15%.

Internships 2018 – 2020
Pentagon Space (2020): Completed 4-month Python Full Stack course; built web applications with Django and SQL.
Teqed Labs (2019): Developed Computer Vision-based attendance system using Deep Learning for face recognition and database management — **Awarded 'Best Project'** for delivering high-impact solution.
Experts Hub (2018): Built machine learning pipeline to classify plant and seedling images with 92% accuracy on 10,000+ image dataset, optimizing species recognition. Led and mentored team of 8 members — **Awarded 'Best Intern'**.

Projects

DAIR-V2X Cooperative 3D Object Detection (PyTorch, CUDA, V2X) 2025
Full reproduction of DAIR-V2X late-fusion model for cooperative 3D object detection in V2X scenarios, achieving 40.01% AP on vehicle class with improved performance metrics: 37.25% 3D AP@0.7 (8.9% gain) and 50.51% BEV AP@0.7 (6.7% gain). Reduced communication cost by 12.3% to 898.10 bytes; implemented LiDAR point cloud processing with visualization tools for 3D and BEV fusion results.
RAG-KnowBot: Personal Knowledge Chatbot (Python, LangChain, Chroma, Llama 3.1) 2025
Built fully local, private Retrieval-Augmented Generation chatbot using Llama 3.1 8B and nomic-embed-text embeddings for interacting with personal documents (PDF, TXT, Markdown).
Features file upload/deletion, customizable prompts, source citations, persistent vector database with Chroma, and hallucination guardrails; runs completely offline for privacy.
Low-Latency Sound Disambiguator (TensorFlow, YAMNet, Streamlit, Mistral) 2025
Real-time sound alert and direction detection system for accessibility; classifies environmental sounds (sirens, barking, alarms, etc) using YAMNet and estimates sound direction via Time Difference of Arrival (TDOA) algorithms.
Optimized for low-latency processing with live microphone input, AI-generated summaries, and web UI featuring visual alerts, confidence trends, and polar compass for directional awareness.
Garbage Segregation System (YOLOv3, CNN, Raspberry Pi, Robotic Arm) Jun. 2020
Designed robotic waste segregation system using YOLOv3 and custom CNNs, achieving 90% garbage classification accuracy with optimized edge inference on Raspberry Pi.
Automated waste sorting and placement with integrated robotic arm; **secured funding from Karnataka State Council for Science & Technology (KSCST)**.

Technical Skills

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Programming Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Shell Scripting
AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv3, CNNs, LangChain, Ollama, ROS2
Deep Learning: Computer Vision, NLP, Reinforcement Learning (SARSA, Q-learning), LLMs (Llama, Mistral), RAG
Specialized Technologies: LiDAR Processing, 3D Object Detection, V2X Communications, CUDA, Embedded Systems (Raspberry Pi)
Tools & Platforms: Docker, Git, Wireshark, Robot Framework, Streamlit, Chroma, Optuna, AWS, JIRA
Databases & Systems: MySQL, SAP, Linux/UNIX, IoT **Networking:** IEEE 802.11mc, DHCPv6, IPv6, Wi-Fi RTT